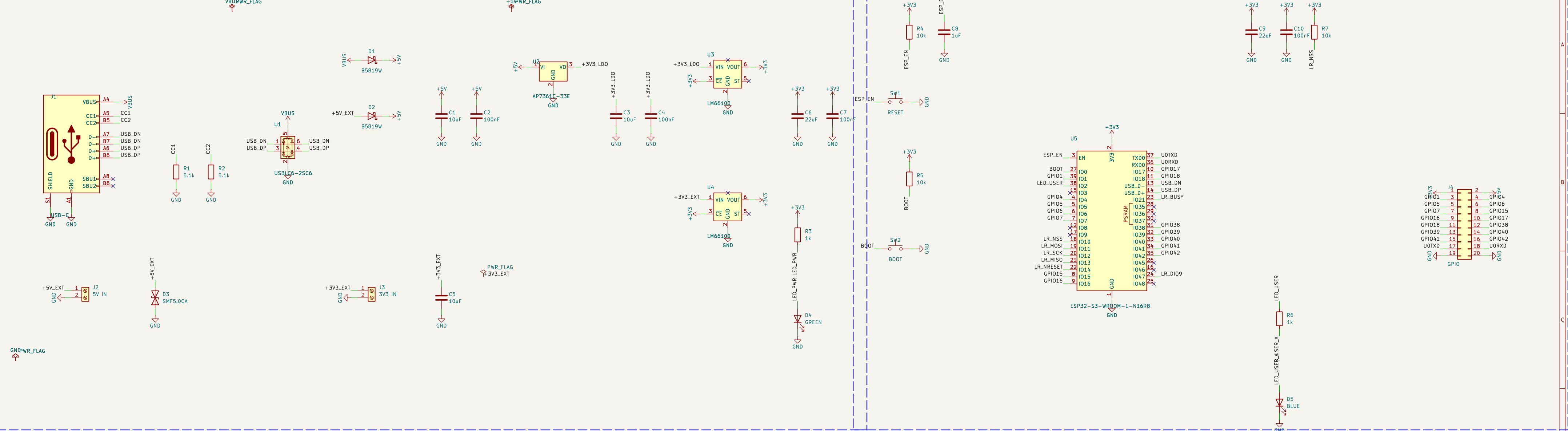
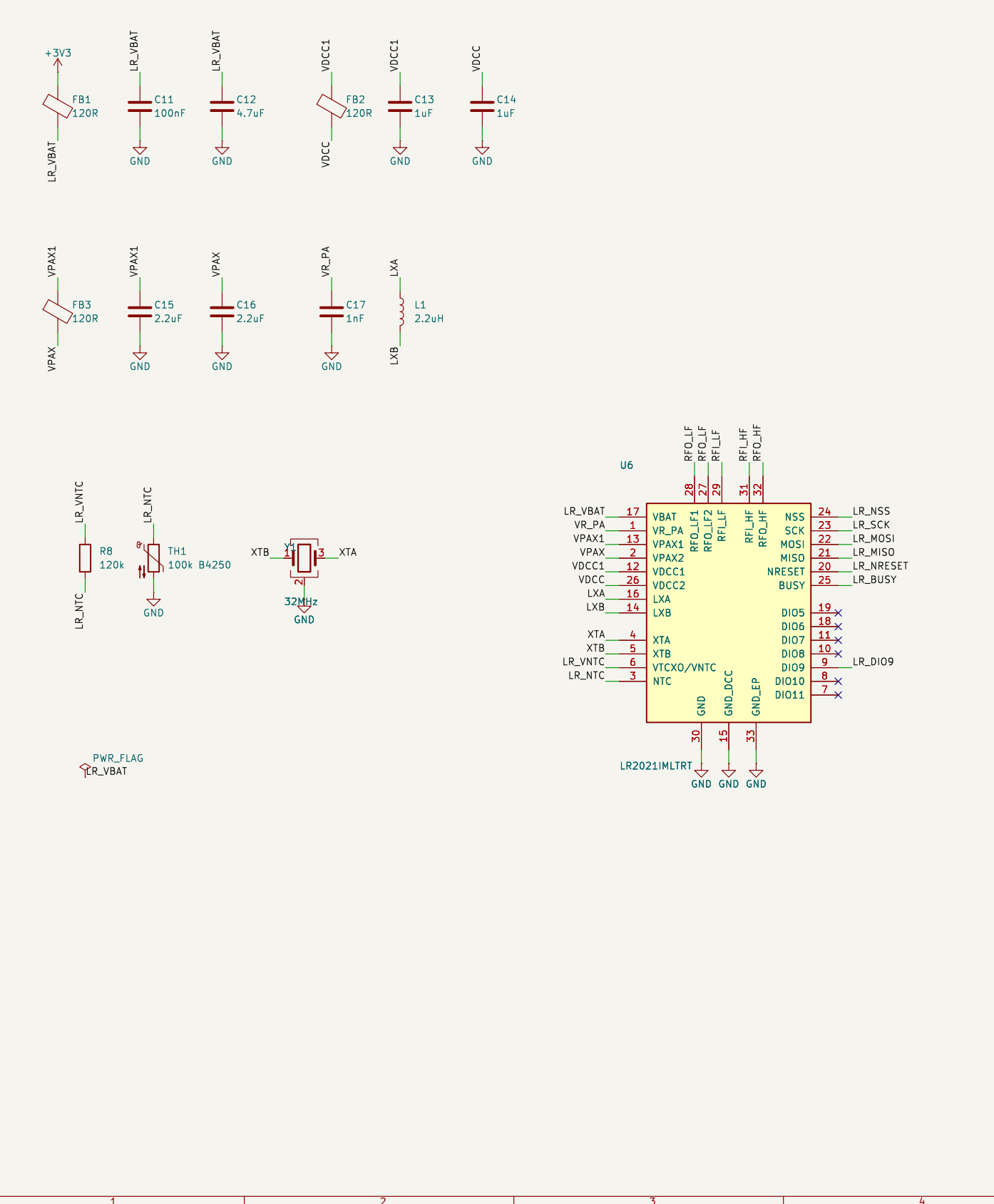


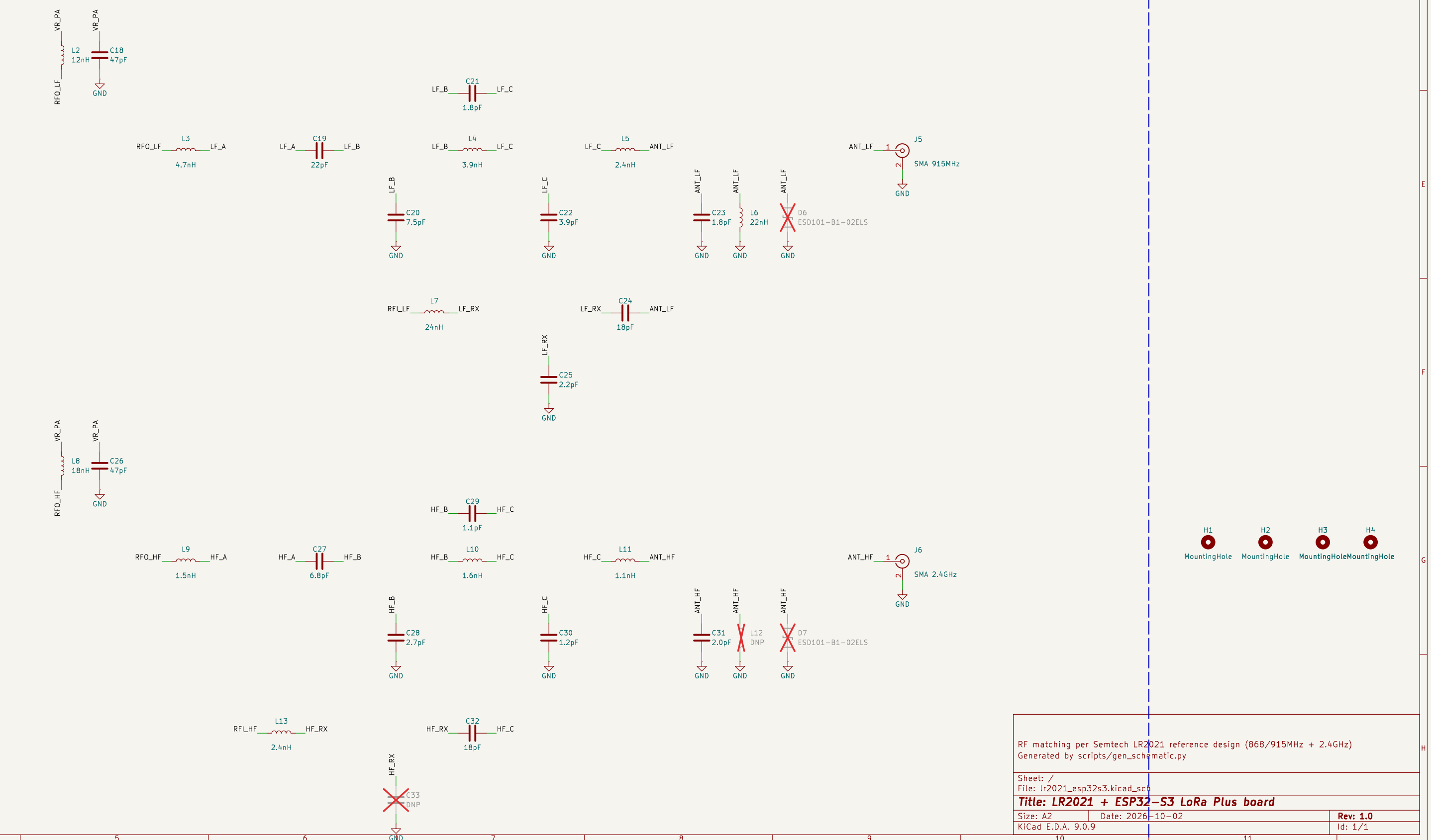
POWER INPUT: USB-C 5V / screw-terminal 5V -> Schottky OR -> AP7361C 1A LDO; external 3.3V and LDO 3.3V combined by two LM66100 ideal diodes (highest source wins, reverse-current blocked) ESP32-S3-WROOM-1-N16R8 (native USB on GPIO19/20; GPIO35-37 used by octal PSRAM)



LR2021 core: SIMO DC-DC, 32MHz crystal with NTC compensation (Semtech reference design)



RF: Semtech LR2021 reference design (switchless direct-tie). Top: 868/915MHz (PA_LF up to +22dBm). Bottom: 2.4GHz (PA_HF up to +12dBm). Keep layout tight, 50 ohm GCPW.



RF matching per Semtech LR2021 reference design (868/915MHz + 2.4GHz)
Generated by scripts/gen_schematic.py